

Module 2

Accounting Statements and Finance

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Why Do We Need Accounting?

1. What do firms own?
 - Real assets (tangible, intangible) and financial assets (cash, securities)

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2. What do firms owe?

Why Do We Need Accounting?

1. What do firms own?
 - Real assets (tangible, intangible) and financial assets (cash, securities)
2. What do firms owe?
3. How much money did firms make?
 - Income? Cash flow?

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?



CNBC, August 18, 2026 (3:50),
“Anthropic tells investors annualized
revenue run rate climbed to \$65
billion in July”

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- We will have a clear understanding after this module



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Three Financial Statements

- **Balance sheet**
 - What do firms own and owe at a point in time



Three Financial Statements

- **Balance sheet**
 - What do firms own and owe at a point in time
- **Income statement**
 - How much did firms earn in the past year



Three Financial Statements

- **Balance sheet**
 - What do firms own and owe at a point in time
- **Income statement**
 - How much did firms earn in the past year
- **Statement of cash flows**
 - Cash inflows and outflows during the past year



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- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure



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- **10-K**, a firm's audited annual report filed with the SEC



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Goals

- Interpret the balance sheet, income statement, and statement of cash flows

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- Distinguish between market and book values
- Explain why income differs from cash flow
- Understand the taxation of corporate and personal income

Module 2, Part 1

The Balance Sheet

Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time

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- Components
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$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

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Where does the balance sheet come from?

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)

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- The numbers stay at the transaction amount, even when the asset is worth more or less later

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today

Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

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3. Build data center			
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- *Should you revise fixed assets if the data center is now worth more than \$2 billion?*

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Breaking Down the Balance Sheet of a Business (\$ Millions)

Current assets

Assets that can generate return within one year

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700
Net tangible fixed assets	28,500

Intangible asset	700
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Total fixed assets	29,200
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Total assets	41,300
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Liabilities and equity

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- Bank loans, debt from banks

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Breaking Down the Balance Sheet of a Business (\$ Millions)

Most liquid	Assets		Liabilities and equity		Paid first
	Current assets		Current liabilities		
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	Accounts receivable	500	Accounts payable	9,900	
	Inventories	9,000	Total current liabilities	10,100	
	Total current assets	12,100	Long-term liabilities	11,300	
	Fixed assets		Total liabilities	21,400	
	Tangible fixed assets		Shareholders' equity		
	Property, plant, equipment	48,200	Paid-in capital	6,300	
	Less accum. depreciation	19,700	Retained earnings	13,600	
	Net tangible fixed assets	28,500	Total shareholders' equity	19,900	
Least liquid	Intangible asset	700			Paid last
	Total fixed assets	29,200			
	Total assets	41,300	Total liabilities and equity	41,300	

- **Liquidity**, the ease of converting an asset into cash

Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
 - Values on balance sheets are book value

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 - E.g., worth \$4 billion today → market value

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 - E.g., the \$2 billion data center built last year → book value, forever
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 - E.g., worth \$4 billion today → market value
- In general, book $\neq$ market value

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- *Should you revise fixed assets if the data center is now worth more than \$2 billion? **No***
- *Should you revise shareholders' equity if someone is willing to buy the business at \$4 billion? **No***

Book Value vs. Market Value

1. Book value of equity = Book value of assets – Book value of liabilities
 - E.g., Apple at the end of fiscal year 2025,
 - Book value of assets, \$359,241 Mn
 - Book value of liabilities, \$285,508 Mn
 - What is the book value of equity?

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1. Book value of equity = Book value of assets – Book value of liabilities
 - E.g., Apple at the end of fiscal year 2025,
 - Book value of assets, \$359,241 Mn
 - Book value of liabilities, \$285,508 Mn
 - What is the book value of equity?
 - Answer, $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$

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 - Answer, $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$
2. Market value of equity = Market price of each share $\times$ Total shares outstanding
 - E.g., Apple at the end of 2025,
 - Share price, about \$271
 - Shares outstanding, 14,800 Mn
 - What is the market value of equity?

Book Value vs. Market Value

1. Book value of equity = Book value of assets – Book value of liabilities
 - E.g., Apple at the end of fiscal year 2025,
 - Book value of assets, \$359,241 Mn
 - Book value of liabilities, \$285,508 Mn
 - What is the book value of equity?
 - Answer, $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$
2. Market value of equity = Market price of each share $\times$ Total shares outstanding
 - E.g., Apple at the end of 2025,
 - Share price, about \$271
 - Shares outstanding, 14,800 Mn
 - What is the market value of equity?
 - Answer, $\$271 \times 14,800 \text{ Mn} = \$4,010,800 \text{ Mn}$

Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm



Book Value vs. Market Value

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- Market value of equity reflects the money investors expect the firm to make in the future



Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future
- Naturally, market value of equity $>$ book value of equity for a successful business



Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?

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- $\$45,000 = \$15,000 + \text{_____} + \$20,000$

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- Which of the following are current assets? Select all that apply.
 - a. Cash.
 - b. A software license (3-year term).
 - c. Wages payable (due in 2 weeks).
 - d. A bank loan (due in 3 years).
 - e. Accounts receivable.
 - f. An office building.

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Recap of Part 1

- Accounting conveys information to investors and stakeholders
- The balance sheet, assets, liabilities, and equity at a point in time
- $\text{Assets} = \text{Liabilities} + \text{Equity}$
- In general, book $\neq$ market value

Module 2, Part 2

The Income Statement

Income Statement

- **Income statement**, how much firms earned in the past year



INCOME STATEMENTS

Income Statement

- **Revenues**, sales made in the period, whether or not cash was collected
 - E.g., a firm sells 100 products to customers, each at a price of \$100
 - $\text{Revenues} = 100 \times \$100 = \$10,000$

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Operating expenses**, the costs of running the business

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Income Statement

- **Operating expenses**, the costs of running the business
 - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
 - E.g., each of the 100 products sold costs the firm \$60 to make
 - $\text{COGS} = 100 \times \$60 = \$6,000$

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- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once



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- Two things happen with depreciation,
 - In the income statement, subtract depreciation from revenues as a non-cash expense



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- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**
- Two things happen with depreciation,
 - In the income statement, subtract depreciation from revenues as a non-cash expense
 - In the **balance sheet**, add to accumulated depreciation



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Cash flow (\$ billions)	-2.0	0	0	0	0	0	
Effect on net income	0	-0.4	-0.4	-0.4	-0.4	-0.4	← Depreciation

Income Statement

- **EBIT**, earnings before interest and taxes
 - $\text{EBIT} = \text{Revenues} - \text{COGS} - \text{SG\&A expenses} - \text{Depreciation}$

	\$ Million
Revenues	80,000
COGS	56,000
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Income Statement

- **Interest expense**, the interest paid on the firm's debt for the period

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Income Statement

- **Taxes**

- Taxable income = EBIT – Interest expense
- Taxes = Taxable income × Tax rate

	\$ Million
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Income Statement

- **Net income**, what the business earned in the period after every expense

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Where Net Income Goes

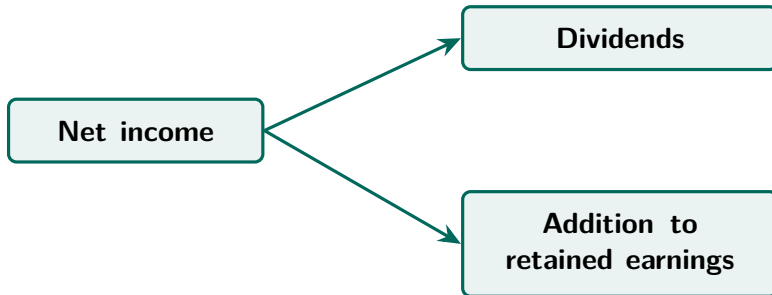
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Where Net Income Goes

- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders

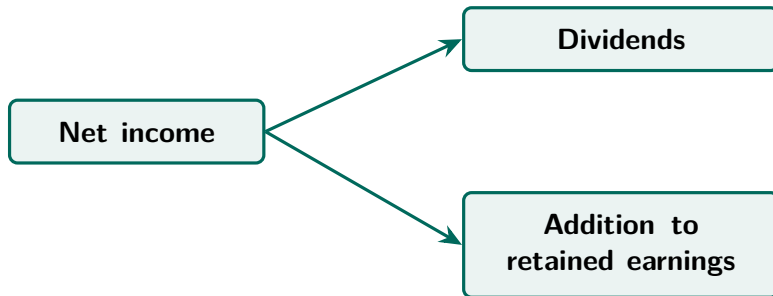
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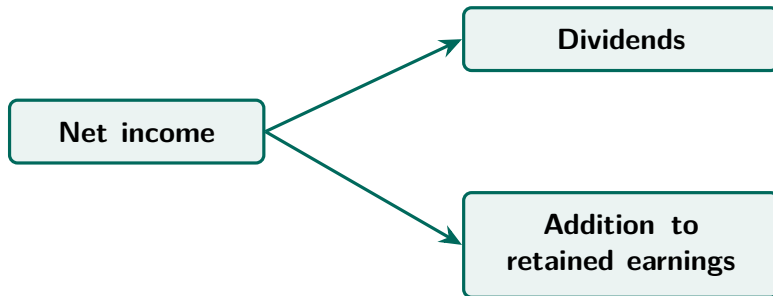
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Where Net Income Goes

- Shareholders have claims to net income
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- **Dividends**, cash paid out to shareholders now
- **Retained earnings**, accumulated profits kept in the business and reinvested on the shareholders' behalf

Income Statement, A Memory Aid

Line item	Letters	Remember it as
Revenues	R	R ainbow
COGS	C	C andy
SG&A expenses	S	S ells
Depreciation	D	D ependably
EBIT	E	E specially
Interest expense	I	I n
Taxable income	T.I.	T exas I ncorporated
Taxes	T	T exas
Net income	N.I.	N ever I ncorporates
Dividends	D	D uring
Addition to retained earnings	R.E.	R e- E lections

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?



CNBC, August 18, 2026 (3:50),
“Anthropic tells investors annualized
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Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- Profitable on an **EBITDA** basis
 - EBITDA, earnings before interest, taxes, depreciation, and amortization
 - Amortization, the same cost-spreading as depreciation, applied to intangible assets



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- Is Anthropic profitable in Q2 2026?
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 - Amortization, the same cost-spreading as depreciation, applied to intangible assets
- Not profitable on a **net income** basis



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Practice Problem

	\$
Revenues	?
COGS	20,000
SG&A expenses	10,000
Depreciation	8,000
EBIT	12,000
Interest expense	?
Taxable income	?
Taxes (tax rate = 21%)	2,100
Net income	7,900
Dividends	2,500
Addition to retained earnings	?

Practice Problem

	\$
Revenues	50,000
COGS	20,000
SG&A expenses	10,000
Depreciation	8,000
EBIT	12,000
Interest expense	2,000
Taxable income	10,000
Taxes (tax rate = 21%)	2,100
Net income	7,900
Dividends	2,500
Addition to retained earnings	5,400

Recap of Part 2

- The income statement, how much firms earned in the past year
- **Depreciation**, this period's share of a fixed asset's cost
- Net income can be kept in the business, or paid out to shareholders
- Taxes are computed **after** depreciation and interest, never before

Module 2, Part 3

The Statement of Cash Flows

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 - **Operating activities**, cash from running the business day to day
 - **Investing activities**, cash spent buying fixed assets and cash received from selling them
 - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

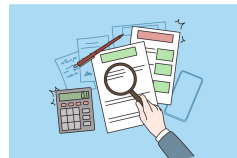
Statement of Cash Flows

Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	The common choice

Direct →



Indirect →



Statement of Cash Flows

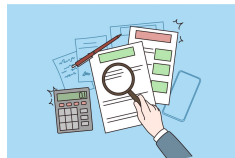
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- Both methods give the same totals

Direct →



Indirect →



Statement of Cash Flows

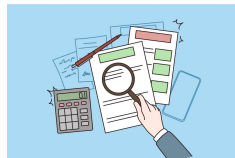
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

Direct →



Indirect →



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Relationship of cash flow (CF) and net income (NI)?

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When spending on fixed assets, $CF = NI - \text{spending on fixed assets}$

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Relationship of cash flow (CF) and net income (NI)?

When spending on fixed assets, $CF = NI - \text{spending on fixed assets}$

When recording depreciation, $CF = NI + \text{depreciation}$

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When raising money, $CF = NI + \text{amount raised}$

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- The firm spends \$60 million producing AI control panels
- None of them have been sold yet



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

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Cash flow (\$ millions)	−60
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<u>Change in inventories</u>	<u>+60</u>



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When adding to inventories, $CF = NI - \text{change in inventories}$



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When firm not yet paid by customers, $CF = NI - \text{change in accounts receivable}$

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When firm pays off suppliers, $CF = NI + \text{change in accounts payable}$



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Operating Activities

Operating activities, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
 - $\text{NWC} = \text{inventories} + \text{accounts receivable} - \text{accounts payable}$
 - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, $CF = NI - \text{change in inventories}$

When firm not yet paid by customers, $CF = NI - \text{change in accounts receivable}$

When firm pays off suppliers, $CF = NI + \text{change in accounts payable}$

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, $CF = NI - \text{change in inventories}$

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Ignoring other adjustments,

$$CF = NI$$

- change in inventories*
- change in accounts receivable*
- + change in accounts payable*

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$$CF = NI - \text{change in (inventories + accounts receivable – accounts payable)}$$

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$$CF = NI - \text{change in (inventories + accounts receivable – accounts payable)}$$

$$CF = NI - \text{change in net working capital (NWC)}$$

Operating Activities

Cash flow from operating activities	
Net income	3,160
Depreciation	3,000
Change in net working capital	<u>-1,000</u>
Cash flow from operating activities	<u>7,160</u>

$$3,160 + 3,000 - (-1,000) = 7,160$$

Investing Activities

Investing activities, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
 - **Capital**, a firm's fixed assets
 - **Capital expenditure**, the amount spent on buying capital during the period
 - Cash flow is *negative* if firms buy capital

Investing Activities

Cash flow from investing activities	
Cash flow from capital expenditure	−3,000
Cash flow from sales of fixed assets	200
Cash flow from investing activities	<u>−2,800</u>

$$-3,000 + 200 = -2,800$$

Financing Activities

Financing activities, cash raised from lenders and shareholders, and cash returned to them

Financing activities show cash flows from,

1. Selling new shares → Cash ↑
2. Paying dividends → Cash ↓
3. Taking new debt, either a loan or a bond → Cash ↑
4. Repaying old debt → Cash ↓

Practice Problem

- During the year, a business sells new shares for \$120,000 and pays dividends of \$30,000. It takes a new loan of \$80,000 and repays \$50,000 of old debt. How much is its cash flow from financing activities?

Practice Problem

- During the year, a business sells new shares for \$120,000 and pays dividends of \$30,000. It takes a new loan of \$80,000 and repays \$50,000 of old debt. How much is its cash flow from financing activities?
- Cash flow from financing activities = $\$120,000 - \$30,000 + \$80,000 - \$50,000 = \$120,000$

Recap of Part 3

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
 - **Operating activities**, cash flow = Net income + Depreciation – Change in NWC
 - **Investing activities**, cash spent buying fixed assets and cash received from selling them
 - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Practice Problem

- Which of the following will make cash flow higher than net income for a firm?
 - a. Increase in inventories.
 - b. Decrease in accounts payable.
 - c. Decrease in accounts receivable.
 - d. Expenditures on fixed assets.

Practice Problem

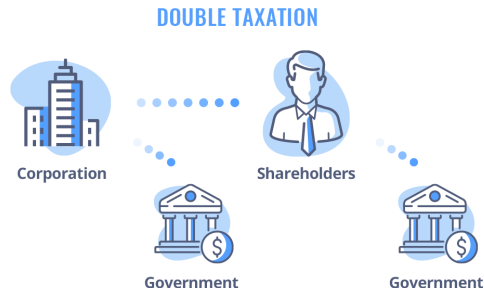
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Module 2, Part 4

Taxation

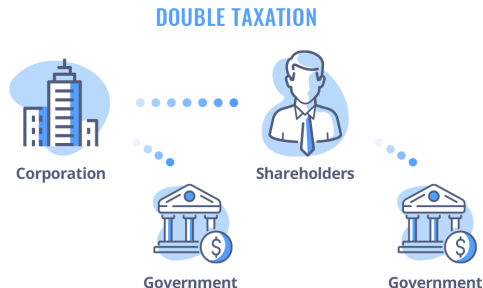
Corporate Taxation, Double Taxation

- Recall from **Module 1**,
 - The corporation pays tax on its income



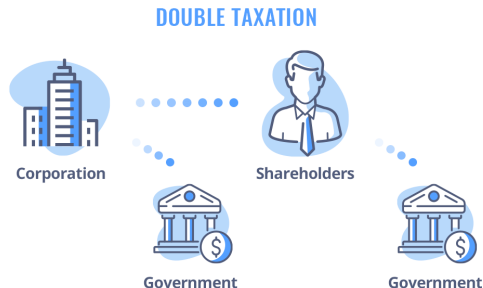
Corporate Taxation, Double Taxation

- Recall from **Module 1**,
 - The corporation pays tax on its income
 - Shareholders pay tax again on what they receive



Corporate Taxation, Double Taxation

- Recall from **Module 1**,
 - The corporation pays tax on its income
 - Shareholders pay tax again on what they receive
- This is **double taxation**



Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is 21%



WSJ, February 28, 2025 (6:57),
“Economists on How Trump’s 2017
Tax Cuts Actually Played Out”

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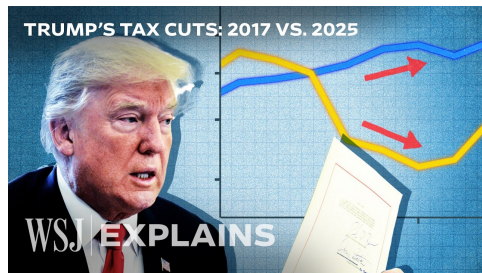
- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic
- **Tax Cuts and Jobs Act (TCJA, 2017)**, cut the rate from 35% to 21%



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Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic
- **Tax Cuts and Jobs Act (TCJA, 2017)**, cut the rate from 35% to 21%
- The tax cut was extended and largely made permanent in 2025



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Tax Shield

- **Tax shield**, the reduction in the taxes a corporation needs to pay that results from a tax-deductible expense



Tax Shield

- **Tax shield**, the reduction in the taxes a corporation needs to pay that results from a tax-deductible expense
- **Tax-deductible expense**, an expense on the income statement that comes before taxable income



Interest Tax Shield

- Interest comes first

Line item	Letters	Remember it as
Revenues	R	R ainbow
COGS	C	C andy
SG&A expenses	S	S ells
Depreciation	D	D ependably
EBIT	E	E specially
Interest expense	I	I n
Taxable income	T.I.	T exas I ncorporated
Taxes	T	T exas
Net income	N.I.	N ever I ncorporates
Dividends	D	D uring
Addition to retained earnings	R.E.	R e- E lections

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Interest Tax Shield

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- Taxes come last
- Interest reduces corporate taxes

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 - act as a “shield” against and reduce corporate taxation

Interest Tax Shield

- Interest expense can
 - act as a “shield” against and reduce corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)

Interest Tax Shield, Example

Corporation A	
EBIT	\$5,000
Interest expense	\$1,000
Taxable income	<u>?</u>
Taxes (tax rate = 21%)	?
Net income	<u>?</u>

Corporation B	
EBIT	\$5,000
Interest expense	\$0
Taxable income	<u>?</u>
Taxes (tax rate = 21%)	?
Net income	<u>?</u>

Interest Tax Shield, Example

Corporation A	
EBIT	\$5,000
Interest expense	\$1,000
Taxable income	\$4,000
Taxes (tax rate = 21%)	\$840
Net income	\$3,160

Corporation B	
EBIT	\$5,000
Interest expense	\$0
Taxable income	?
Taxes (tax rate = 21%)	?
Net income	?

Interest Tax Shield, Example

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EBIT	\$5,000
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Taxable income	\$4,000
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Corporation B	
EBIT	\$5,000
Interest expense	\$0
Taxable income	\$5,000
Taxes (tax rate = 21%)	\$1,050
Net income	\$3,950

Interest Tax Shield, Example

Corporation A

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Interest expense	\$1,000	→ debt investors
Taxable income	\$4,000	
Taxes (tax rate = 21%)	\$840	
Net income	\$3,160	

Corporation B

EBIT	\$5,000	
Interest expense	\$0	→ debt investors
Taxable income	\$5,000	
Taxes (tax rate = 21%)	\$1,050	
Net income	\$3,950	

Interest Tax Shield, Example

Corporation A

EBIT	\$5,000	
Interest expense	\$1,000	→ debt investors
Taxable income	\$4,000	
Taxes (tax rate = 21%)	\$840	
Net income	\$3,160	→ equity investors

Corporation B

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Interest expense	\$0	→ debt investors
Taxable income	\$5,000	
Taxes (tax rate = 21%)	\$1,050	
Net income	\$3,950	→ equity investors

Interest Tax Shield, Example

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Corporation B

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Interest Tax Shield, Example

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Taxes (tax rate = 21%)	\$1,050		
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Investors get \$4,160 in total with debt, \$3,950 without

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Taxes (tax rate = 21%)	\$840
Net income	\$3,160 → equity investors

} **\$4,160**

Corporation B	
EBIT	\$5,000
Interest expense	\$0 → debt investors
Taxable income	\$5,000
Taxes (tax rate = 21%)	\$1,050
Net income	\$3,950 → equity investors

} **\$3,950**

Investors get \$4,160 in total with debt, \$3,950 without

Investors get \$210 more in total with debt

Interest Tax Shield

- Interest expense can
 - act as a “shield” against and reduce corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)
- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

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- In the example, $21\% \times \$1,000 = \210
- Should a corporation take on as much debt as possible? No, bankruptcy costs.

Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1

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- So taxes fall by the tax rate $(21\%) \times \$1 = \text{¢}21$

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$$\underbrace{\text{Total CF to investors}}_{\uparrow \text{ by } \text{¢}21} = \underbrace{\text{Debt investors}}_{\uparrow \text{ by } \$1} + \underbrace{\text{Equity investors}}_{\downarrow \text{ by } \text{¢}79}$$

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- Each dollar of interest raises the *total* cash flow to investors by ¢21

Practice Problem

- A corporation pays a 21% tax rate. It takes on debt and pays \$1 more in interest each year. In that year, how does the cash flow to each investor group change?
 - a. Debt investors get \$1 more, equity investors get \$1 less, no change in total
 - b. Debt investors get \$1 more, equity investors get \$0.79 less, \$0.21 more in total
 - c. Debt investors get \$0.79 more, equity investors get \$1 less, \$0.21 less in total
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 - c. Debt investors get \$0.79 more, equity investors get \$1 less, \$0.21 less in total
 - d. Debt investors get \$1 more, equity investors get \$0.21 less, \$0.79 more in total
- Net income falls by $(1 - 21\%) \times \$1 = \0.79 , and the interest tax shield is $21\% \times \$1 = \0.21

Depreciation Tax Shield

- Depreciation comes first

Line item	Letters	Remember it as
Revenues	R	R ainbow
COGS	C	C andy
SG&A expenses	S	S ells
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Depreciation Tax Shield, Example

- Example, Suppose a corporation has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

Income Statement (\$ Million)	
	Baseline
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160

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Interest expense	1,000	1,000
Taxable income	4,000	3,000
Taxes	840	630
Net income	3,160	

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Taxable income	4,000	3,000
Taxes	840	630
Net income	3,160	2,370

Depreciation Tax Shield, Example

- Net income *falls* by \$790 million

\$ Million	Baseline	More depreciation	Change
Net income	3,160	2,370	−790
Add back depreciation	_____	_____	_____
Cash flow	_____	_____	_____

Depreciation Tax Shield, Example

- Net income *falls* by \$790 million
- Recall Part 3,
 $\text{CF from operating activities} = \text{NI} + \text{Depreciation} - \text{Change in NWC}$

\$ Million	Baseline	More depreciation	Change
Net income	3,160	2,370	-790
Add back depreciation	3,000	4,000	+1,000
Cash flow			

Depreciation Tax Shield, Example

- Net income *falls* by \$790 million
- Recall Part 3,
 $\text{CF from operating activities} = \text{NI} + \text{Depreciation} - \text{Change in NWC}$
- Equity investors have claims to cash flow from operating activities

\$ Million	Baseline	More depreciation	Change
Net income	3,160	2,370	-790
Add back depreciation	3,000	4,000	+1,000
Cash flow			

Depreciation Tax Shield, Example

- Net income *falls* by \$790 million
- Recall Part 3,
 $\text{CF from operating activities} = \text{NI} + \text{Depreciation} - \text{Change in NWC}$
- Equity investors have claims to cash flow from operating activities
- Cash flow *rises* by \$210 million

\$ Million	Baseline	More depreciation	Change
Net income	3,160	2,370	-790
Add back depreciation	3,000	4,000	+1,000
Cash flow	6,160	6,370	+210

Depreciation Tax Shield

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 - act as a “shield” against and reduce corporate taxation
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 - increase the *total* cash flow to investors (debt and equity combined)
- **Depreciation tax shield**, the tax saving that depreciation creates, as depreciation is deducted (subtracted) before taxes

$$\text{Depreciation tax shield} = \text{Tax rate} \times \text{Depreciation}$$

- In the example, $21\% \times \$1,000 = \210
- Can a corporation have as much depreciation as possible? No, constrained by cost of acquisition and accounting rules.

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- Equity investors get ¢21 more
- Each dollar of depreciation raises the *total* cash flow to investors by ¢21

Practice Problem

- A corporation pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its cash flow?

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- Taxes fall by $21\% \times \$200 \text{ million} = \42 million
- Cash flow rises by the same \$42 million, the depreciation tax shield

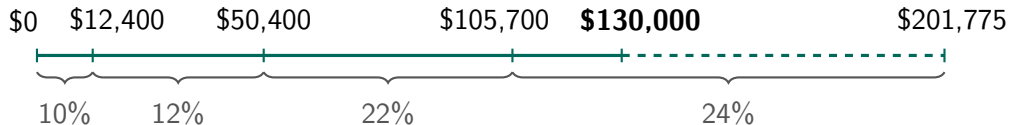
Personal Taxation

- Personal tax rates matter for salaried employees and non-corporate businesses
- **Marginal tax rate**, the tax rate applied to each extra dollar of income
- **Average tax rate**, total taxes / total income

Personal Taxation

Taxable Income (Single Taxpayers, 2026)	Marginal Tax Rate (%)
\$0–\$12,400	10
\$12,401–\$50,400	12
\$50,401–\$105,700	22
\$105,701–\$201,775	24
\$201,776–\$256,225	32
\$256,226–\$640,600	35
\$640,601 and above	37

Source, [IRS Rev. Proc. 2025-32](#)



Example, Average Tax Rate

Suppose your salary is \$130,000 per year. What would be your average tax rate?

Taxable Income (Single Taxpayers, 2026)	Marginal Tax Rate (%)
\$0–\$12,400	10
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- Total tax = $\$12,400 \times 10\% + (\$50,400 - \$12,400) \times 12\% + (\$105,700 - \$50,400) \times 22\% + (\$130,000 - \$105,700) \times 24\% = \$23,798$
- Average tax rate = $23,798 / 130,000 = 18.31\%$

Practice Problem

- Suppose your salary is \$30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

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- Average tax rate = $3,352 / 30,000 = 11.17\%$

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- Average tax rate = $3,352 / 30,000 = 11.17\%$
- Marginal tax rate = 12%

Recap of Part 4

- The U.S. federal corporate tax rate is **21%**
- Interest and depreciation are deducted (subtracted) before taxes, so both shield income from tax
- A tax shield is worth the tax rate $\times$ the deduction, raising the *total* cash flow to investors
- Marginal rate applies to each extra dollar of income, average rate is total taxes over total income

Key Takeaways

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- Interest and depreciation shield taxes, raising the *total* cash flow to investors

Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?

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- $\text{Taxable income} = \text{Taxes} / \text{Tax rate} = \$2,100 / 21\% = \$10,000$

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- Taxable income = Taxes / Tax rate = $\$2,100 / 21\% = \$10,000$
- Net income = Taxable income – Taxes = $\$10,000 - \$2,100 = \$7,900$

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- EBIT = Taxable income + Interest expense = $\$10,000 + \$2,000 = \$12,000$

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- Net income = Taxable income – Taxes = $\$10,000 - \$2,100 = \$7,900$
- EBIT = Taxable income + Interest expense = $\$10,000 + \$2,000 = \$12,000$
- Revenues = EBIT + COGS + SG&A expenses + Depreciation = $\$12,000 + \$20,000 + \$10,000 + \$8,000 = \$50,000$

Practice Problem

If a firm's net income is positive and its depreciation is positive, which of the following could account for a negative cash flow from operating activities?

- a. Inventories decrease while accounts receivable and accounts payable stay the same.
- b. Accounts receivable increases while inventories and accounts payable stay the same.
- c. The firm raises money by issuing stocks.
- d. A large addition is made to plant and equipment.

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Thank you!

Please let me know if you have any questions.